

Andrew Chang

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EDUCATION

University of California, Irvine

Irvine, CA

BS in Computer Science, Minor in Statistics & Informatics

Expected Graduation: December 2026

Coursework: Data Structures & Algorithms, Parallel and Distributed Computing, Databases, Machine Learning, Object Oriented Programming, Compilers & Interpreters, Computer Networks, Operating Systems

EXPERIENCE

SoFi

June 2025 – September 2025

Software Engineer Intern

San Francisco, CA

- Developed a **production-grade monitoring system for external vendors** in **Go (FX framework)** with PostgreSQL, deployed on **AWS EC2** via **Kubernetes** and **Argo CD**, cutting detection time by 60% and preventing \$100K+ in downtime costs
- Designed and configured dynamic **Datadog dashboards** to visualize latency, error rates, and SLA breaches, enabling **30+ engineers** to pinpoint faulty integrations quickly and reducing manual log investigation by 50%
- Instrumented multiple RESTful endpoints with the **OpenTelemetry SDK**, adding context propagation and trace/span attribution across services, increasing observability coverage by **40%**

Tesla

December 2024 – May 2025

Software Engineer Intern

Fremont, CA

- Led the **full-stack** development of a factory analytics dashboard using **React/TypeScript** and **Go**, reducing Megapack production-line defects by 25% and **savings over \$500,000 annually** in scrap and downtime costs
- Engineered **Go REST APIs** and backend logic to collect, validate, and stream Megapack factory metrics into **Prometheus and Kafka**, enabling event-driven monitoring across multiple production lines
- Architected a multi-section analytics UI, managing complex component state, conditional rendering paths, and data-dependent panels with **React Hook Form** and **React Query** for synchronization with factory metrics APIs

Ingram Micro

June 2024 – December 2024

Software Engineer Intern

Irvine, CA

- Integrated **5+ new data sources** into **Elasticsearch** using **Java** and **Spring Boot** within the **Kraken** platform, increasing indexed product coverage by **40%** and improving search reliability for **1000+ daily users**
- Deployed and maintained **Elasticsearch pipelines**, implementing automated testing with **JUnit** and **Mockito** along with validation scripts to ensure **data consistency**, reduce **indexing errors**, and improve **production search reliability**

San Antonio Spurs

September 2023 – May 2024

Data Engineer Intern

San Antonio, TX

- Built and maintained **10+ ETL pipelines** using **Python** and **Apache Airflow** to extract, transform, and load **customer marketing data** from external APIs into **AWS Redshift**, improving data freshness and increasing processing speed by **200%**
- Served **10+ API endpoints** supporting **5+ engineers**, improving access to **customer marketing data** and accelerating development of internal dashboards and analytics pipelines

PROJECTS

Sentinel-RPC | *Distributed Rate Limiting Service*

- Developed a high-performance rate limiting service in **Java** using **gRPC** and **Redis**, implementing the Token Bucket algorithm to manage traffic for microservices at 10k+ requests per second
- Engineered thread-safe request processing using **Java Concurrency** utilities and **Lettuce** for asynchronous Redis I/O, ensuring consistent sub-5ms latency across distributed nodes

SKILLS

Languages: Go, Python, C++, Java, SQL, JavaScript, TypeScript, Swift

Frameworks & Libraries: React, Next.js, Node.js, Django, Spring Boot, FastAPI, SwiftUI

Tools & Platforms: PostgreSQL, MySQL, MongoDB, Redis, Docker, Kubernetes, AWS, GCP, Kafka, Airflow, Datadog